SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5109

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 2 **Total Marks:100**

Annexure A

CONCEPT MAPPING

Code of Conduct:

I, K.Mounith(192425090)certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Mounith

Annexure A

CONCEPT MAPPING

1. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
    a. Show relationships between these concepts
    b. Analyze how each contributes to secure communication
    c. Include one real-world application for each

2. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected
b. Map their role in encryption/decryption
c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

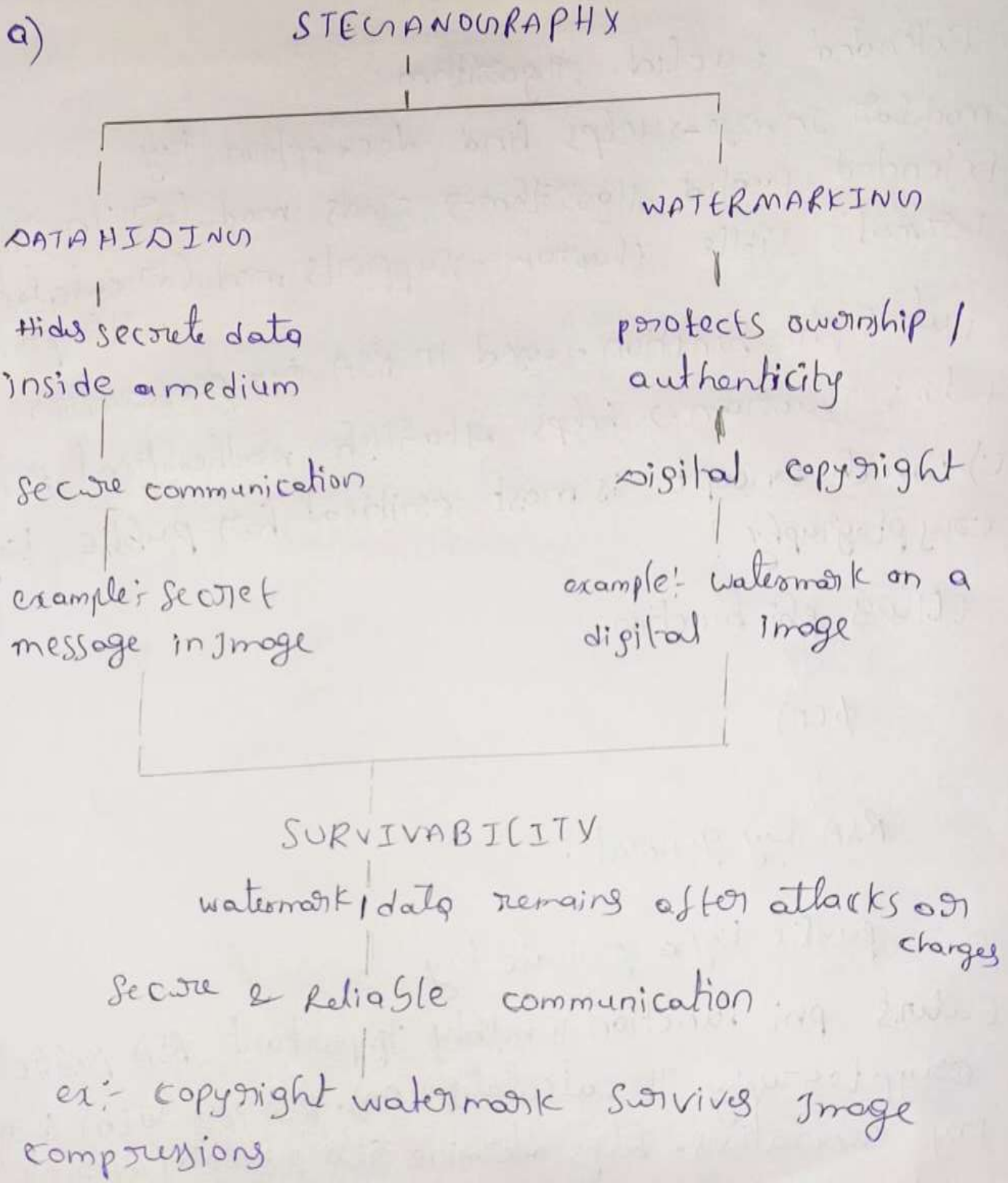
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, wellstructured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

8-10-26

CO1 - Assessment 2

1) concept map - Steganography.

a)



CSA 5109 - Cryptography Slot-A

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5) Analyze how each contributes to secure communication

- Data Hiding → keeps secret information hidden
- watermarking → provides ownership and authenticity.
- Survivability → keeps hidden data safe after modifications or attacks.
- Together → Improve confidentiality, authecity and reliability.

6) Include one real-world application for each:

Concept	Real-world application
Data Hiding	Secret message hidden in an Image.
watermarking	copyright watermark on a photo.
Survivability	watermark remaining after image compression

2. Number theory concepts is mathematically cryptography

a) Show how each concept is mathematically connected

Extended Euclid Algorithm

↓

modular Inverse

↓

Key calculation

↑

Euler phi function

↓

Euler Theorem

↓

encryption/decryption

Fermat's little theorem

↓

Modular Arithmetic

↓

cryptographic calculations.

b) map their role in encryption / decryption

Euler phi function

↓

calculate $\phi(n)$

↓

Key Generation

↓

encryption → decryption.

modular Inverse

Extended Euclid Algorithm.

- modular Inverse → Helps find decryption key.
- Extended Euclid Algorithm → Finds modular Inverse.
- Formal's little theorem → supports modular calculations with primes.
- Euler's phi function → used in RSA key generation
- Euler's Theorem → helps establish mathematical RSA.

c) which concept is most critical for public key cryptography?

Euler's phi function.

↓

$\phi(n)$

↓

RSA key generation

↓

public key ⇔ private key

Euler's phi function is highly important RSA public-key cryptography. It calculates $\psi(n)$. RSA uses $\psi(n)$ during key generations. Helps determine b/w public key & private key.